

Arrays

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Agenda

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Arrays

Objectives

At the end of this module, you will be able to :

- Work with one dimensional and two dimensional arrays in Java

Arrays



Arrays

- An array is a container object that holds a fixed number of values of a single type
- When an array is created, the length of an array is fixed
- Array elements are automatically initialized with the default value of their type, When an array is created
- Array can be created using the **new** keyword

Ex:

```
int[] x = new int[5]; // defining an integer array for 5
```

blocks

Arrays (Contd.).

- Alternatively, we can create and initialize array as below format

```
int[] x = {10, 20, 30};
```

```
int[] x = new int[]{10, 20, 30};
```

- Here the length of an array is determined by the number of values provided between { and }
- The built-in length property determines the size of any array

Ex:

```
int[] x = new int[10];
```

```
int x_len = x.length;
```

Array - Example

```
public class ArrayDemo {  
    public static void main(String[] args) {  
        int[] x; // declares an array of integers  
        x = new int[5]; // allocates memory for 5 integers  
        x[0] = 11;  
        X[4] = 22;  
        System.out.println("Element at index 0: " + x[0]);  
        System.out.println("Element at index 1: " + x[1]);  
        System.out.println("Element at index 4: " + x[4]);  
    }  
}
```

Output:

Element at index 0: 11

Element at index 1: 0

Element at index 4: 22

Array Bounds, Array Resizing

- Array subscripts begin with 0
- Can't access an array element beyond the range
- Can't resize an array. Can use the same reference variable to refer new array

```
int x[] = new int [5];  
x= new int [10];
```


Array copy

- To copy array elements from one array to another array, we can use arraycopy static method from System class

- Syntax:

```
public static void arraycopy(Object s,int  
                             sIndex,Object d,int dIndex,int lngth)
```

- Ex:

```
int source[] = {1, 2, 3, 4, 5, 6};
```

```
int dest[] = new int[10];
```

```
System.arraycopy(source,0, dest,0,source.length);
```

Array Copy - Example

```
public class ArrayLengthDemo {  
    public static void main(String[] args) {  
        // creates and initializes an array of integers  
        int[] source = {100, 200, 300};  
        // creates an integer array with 3 element  
        int[] dest = new int[source.length];  
        // copying an elements from source to dest array  
        System.arraycopy(source, 0, dest, 0, source.length);  
        for (int i = 0; i < dest.length; i++)  
            System.out.println("Element at index " + i + ": " +  
                               dest[i]);  
    }  
}
```

Output:

Element at index 0: 100
Element at index 1: 200
Element at index 3: 300

Two-Dimensional Arrays

- Two-dimensional arrays are arrays of arrays
- Initializing two-dimensional arrays:

```
int[][] y = new int[3][3];
```

The 1st dimension represent rows or number of one dimension, the 2nd dimension represent columns or number of elements in the each one dimensions

- The curly braces { } may also be used to initialize two dimensional arrays
- Ex:

```
int[][] y = { {1,2,3}, {4,5,6}, {7,8,9} };
```

```
int[][] y = new int[3][] { {1,2,3}, {4,5}, {7,8,9} };
```

Two-Dimensional Arrays (Contd.).

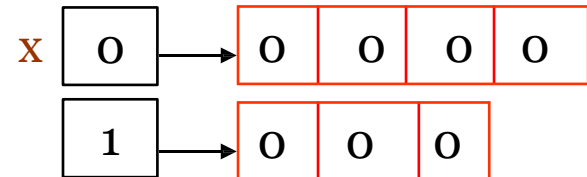
- You can initialize the row dimension without initializing the columns but not vice versa

```
int[][] x = new int[3][];
```

```
int[][] x = new int[][3]; //error
```

- The length of the columns can vary for each row and initialize number of columns in each row
- Ex1:

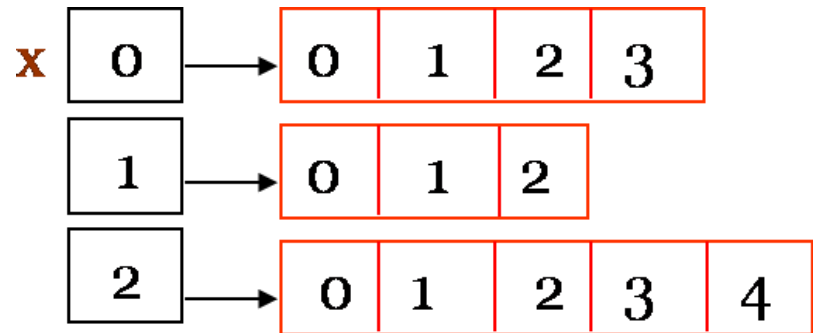
```
int [][]x = new int [2][];  
x[0] = new int[5];  
x[1] = new int [3];
```



Two-Dimensional Arrays (Contd.).

Ex2:

```
int [][]x = new int [3][];  
x[0] = new int[] {0,1,2,3};  
x[1] = new int [] {0,1,2};  
x[2] = new  
int[] {0,1,2,3,4};
```



Two-Dimensional Array - Example

```
/* Program to under stand two-dimensional arrays */
```

```
class TwoDimDemo {  
    public static void main(String[] args) {  
        int [][] x = new int[3][]; // initialize number of rows  
        x[0] = new int[3]; // define number of columns in each row  
        x[1] = new int[2];  
        x[2] = new int[5];  
        for(int i=0; i < x.length; i++) { // print array elements  
            for (int j=0; j < x[i].length; j++) {  
                x[i][j] = i;  
                System.out.print(x[i][j]);  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:
000
11
22222

Quiz

- Select which of the following are valid array definition
1. `int[] a;`
`a = new int[5];`
 2. `int a[] = new int[5]`
 3. `int a[5] = new int[5];`
 4. `int a[] = {1,2,3};`
 5. `int[] a = new int[]{1,2,3};`
 6. `int[] a = new int[5]{1,2,3,4};`

Quiz (Contd.).

What will be the result, if we try to compile and execute the following codes

```
class Sample {  
    public static void main(String[] args) {  
        int[] a = new int[5]{1,2,3};  
        for(int i : a)  
            System.out.println(i);  
    }  
}
```


Quiz

What will be the result, if we try to compile and execute the following codes

```
1. class Sample {  
    public static void main(String[] args) {  
        while(false)  
            System.out.println("while loop");  
    }  
}
```

```
2. class Sample {  
    public static void main(String[] args) {  
        for( ; ; )  
            System.out.println("For loop");  
    }  
}
```

Quiz

What will be the result, if we try to compile and execute the following code?

```
class Test {  
    public static void main(String [ ] args) {  
        int [ ] x=new int[10];  
        System.out.println(x[4]);  
    }  
}
```

Quiz

What will be the result, if we try to compile and execute the following code?

```
class Test {  
    public static void main(String [ ] args) {  
        int x[ ][ ]=new int[10][ ];  
        System.out.println(x[4][0]);  
    }  
}
```



Summary

In this session, you were able to :

- Understand how to work with single and two dimensional arrays in Java

Thank You

